Website analysis of codeforces

Overview:

Codeforces is a competitive programming platform that hosts regular contests, provides a vast problem archive, and fosters a vibrant community of programmers. It is widely used by individuals preparing for coding interviews, competitive programming contests, and algorithmic challenges.

UI/User panel:

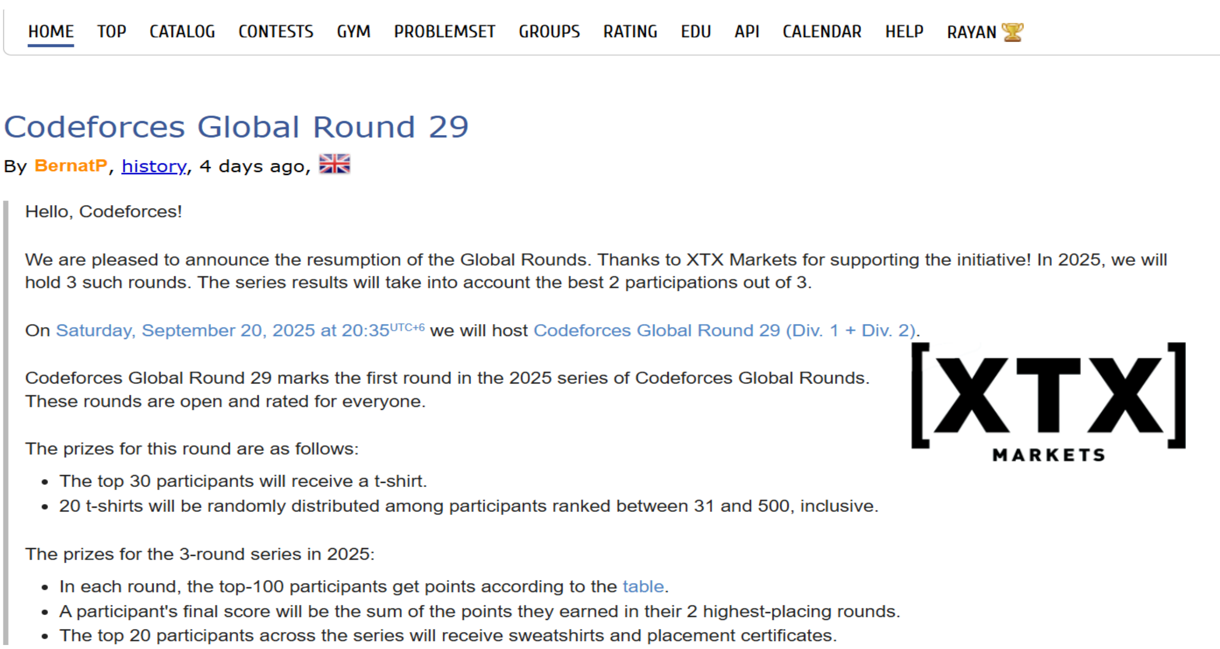
Users need to register and login to enter into the website.

Homepage

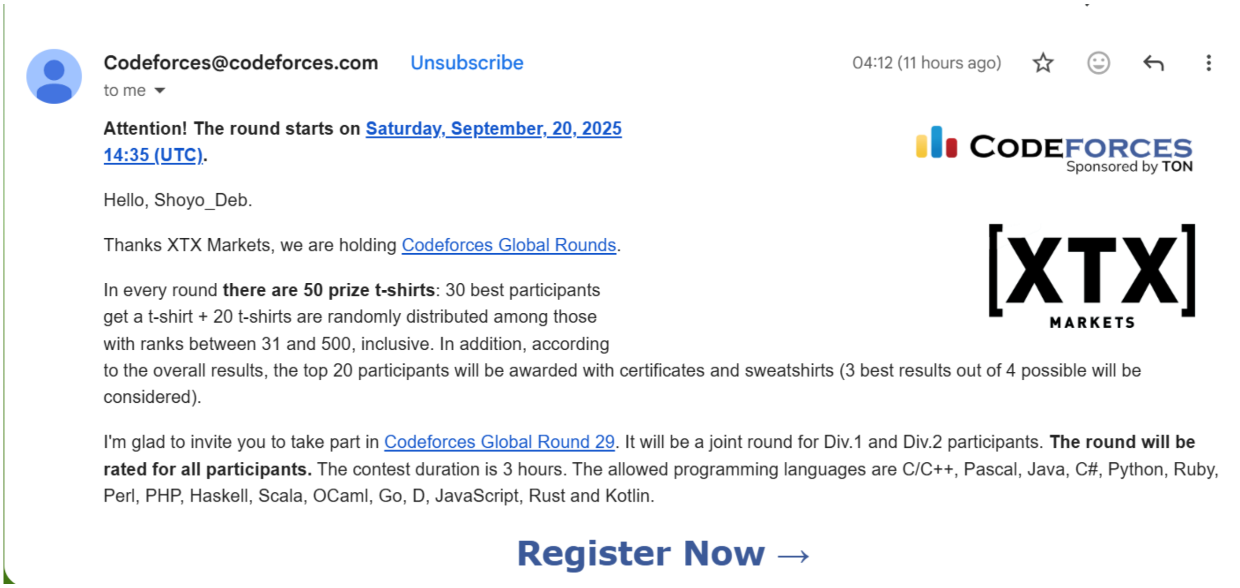
Codeforces homepage provides a quick overview of upcoming contests, recent problems, announcements, and blogs. Users can navigate to contests, problem sets, and community discussions from the top menu.

Observations:

- Important features like “Upcoming Contests” and “Recent Blogs” are easy to locate.



Send emails to users to register in the upcoming contest:



Codeforces@codeforces.com Unsubscribe

to me

04:12 (11 hours ago)

Attention! The round starts on [Saturday, September, 20, 2025 14:35 (UTC)].

CODEFORCES Sponsored by TON

Hello, Shoyo_Deb.

Thanks XTX Markets, we are holding Codeforces Global Rounds.

[XTX] MARKETS

In every round **there are 50 prize t-shirts**: 30 best participants get a t-shirt + 20 t-shirts are randomly distributed among those with ranks between 31 and 500, inclusive. In addition, according to the overall results, the top 20 participants will be awarded with certificates and sweatshirts (3 best results out of 4 possible will be considered).

I'm glad to invite you to take part in Codeforces Global Round 29. It will be a joint round for Div.1 and Div.2 participants. **The round will be rated for all participants.** The contest duration is 3 hours. The allowed programming languages are C/C++, Pascal, Java, C#, Python, Ruby, Perl, PHP, Haskell, Scala, OCaml, Go, D, JavaScript, Rust and Kotlin.

Register Now →

2. Contest Page

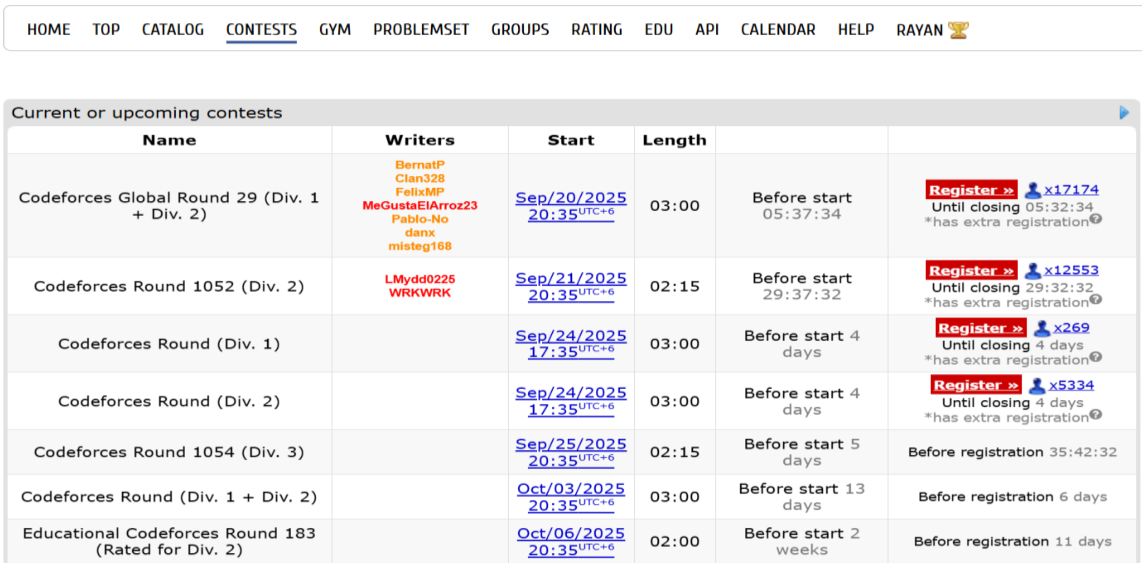
Contest System

- **Regular Contests**: Codeforces organizes contests approximately once a week, categorized into different divisions based on user ratings. This division system allows participants to compete against others of similar skill levels.
- **Educational Rounds**: These are special contests designed to teach specific algorithms and data structures, often accompanied by tutorials and editorials.
- **Hacking Phase**: After a contest, participants can challenge others' solutions in a 12-hour hacking phase, promoting deeper understanding and learning.

Description:

- Shows active and upcoming contests with division information (Div 1, Div 2, Div 3).
- Clicking a contest displays problem list, standings, and contest rules**.**

- Editorials and discussion forums are available for most problems.



HOME TOP CATALOG CONTESTS GYM PROBLEMSET GROUPS RATING EDU API CALENDAR HELP RAYAN

Current or upcoming contests

Name	Writers	Start	Length		
Codeforces Global Round 29 (Div. 1 + Div. 2)	BernatP Clan328 FelixMP MeGustaElArroz23 Pablo-No danx misteg168	Sep/20/2025 20:35 UTC+6	03:00	Before start 05:37:34	Register » x17174 Until closing 05:32:34 *has extra registration
Codeforces Round 1052 (Div. 2)	LMydd0225 WRKWRK	Sep/21/2025 20:35 UTC+6	02:15	Before start 29:37:32	Register » x12553 Until closing 29:32:32 *has extra registration
Codeforces Round (Div. 1)		Sep/24/2025 17:35 UTC+6	03:00	Before start 4 days	Register » x269 Until closing 4 days *has extra registration
Codeforces Round (Div. 2)		Sep/24/2025 17:35 UTC+6	03:00	Before start 4 days	Register » x5334 Until closing 4 days *has extra registration
Codeforces Round 1054 (Div. 3)		Sep/25/2025 20:35 UTC+6	02:15	Before start 5 days	Before registration 35:42:32
Codeforces Round (Div. 1 + Div. 2)		Oct/03/2025 20:35 UTC+6	03:00	Before start 13 days	Before registration 6 days
Educational Codeforces Round 183 (Rated for Div. 2)		Oct/06/2025 20:35 UTC+6	02:00	Before start 2 weeks	Before registration 11 days

3. Problem Page

Problem Set and Difficulty Levels

- **Problem Catalog**: Codeforces offers a vast collection of problems categorized by difficulty, tags (e.g., dynamic programming, graphs), and rating. This allows users to practice specific topics and track their progress.
- **Dynamic Difficulty Adjustment**: The platform adjusts problem difficulty based on the user's rating, ensuring a tailored experience that promotes continuous learning.

Description:

- Each problem includes statement, constraints, input/output format, examples, and tags.

- Users can submit solutions in multiple programming languages.

MAIN | ACMSGURU | PROBLEMS | SUBMIT | STATUS | STANDINGS | CUSTOM TEST

Problems

#	Name	Tags	Difficulty	Solved
2148G	Farmer John's Last Wish	binary search, data structures, math, number theory	1900	x2297
2148F	Gravity Falls	greedy, implementation, sortings	1800	x3411
2148E	Split	binary search, data structures, two pointers	1200	x8946
2148D	Destruction of the Dandelion Fields	constructive algorithms, greedy, sortings	1000	x17074
2148C	Pacer	greedy, math	900	x17416
2148B	Lasers	geometry	800	x23136
2148A	Sublime Sequence	brute force, hashing, math	800	x29891

This is the easy version of the problem. The difference between the versions is that in this version, $n \le 300$. You can hack only if you solved all versions of this problem.

A sequence $b_1, b_2, \ldots, b_k$ is called **good** if there exists a coloring of each index i in red or blue such that for every pair of indices $i < j$ with $b_i > b_j$, the colors assigned to i and j are different.

You are given a sequence $a_1, a_2, \ldots, a_n$. Compute the number of good subsequences of the sequence, including the empty subsequence *. Since the answer can be very large, output it modulo $10^9 + 7$.

*A sequence b is a subsequence of a sequence a if b can be obtained from a by the deletion of several (possibly, zero or all) element from arbitrary positions.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 100$). The description of the test cases follows.

The first line contains an integer n ($1 \le n \le 300$) — the length of the sequence

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$) — the contents of the sequence.

It is guaranteed that the sum of n over all test cases does not exceed 300.

Output

For each test case, output a single line containing the number of good subsequences modulo $10^9 + 7$.

Example

input

```
4
4
4 2 3 1
7
7 6 1 2 3 3 2
5
1 1 1 1 1
11
7 2 1 9 7 3 4 1 3 5 3
```

output

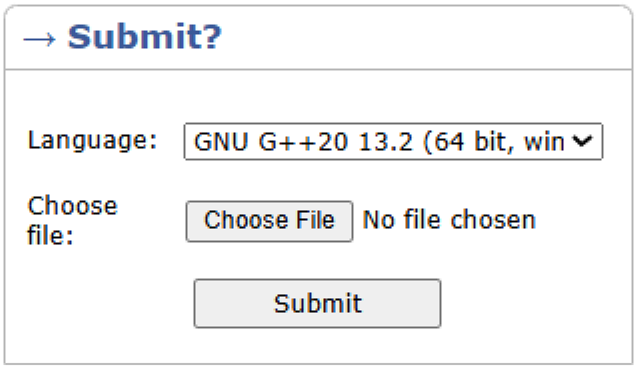
```
13
73
32
619
```

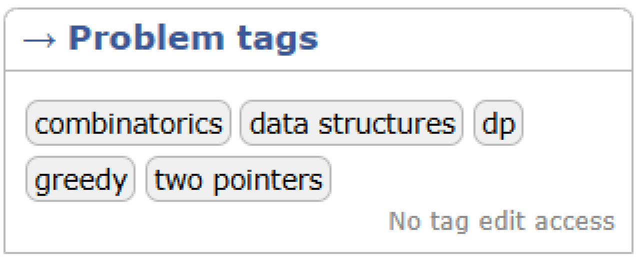
Note

In the first test case, the subsequences that are **not** good are $[4, 3, 1]$, $[4, 2, 1]$, and $[4, 2, 3, 1]$. Since there are 16 subsequences in total, this means there are $16 - 3 = 13$ good subsequences.

In the third test case, every subsequence is good.

Submission option and tag for problems:

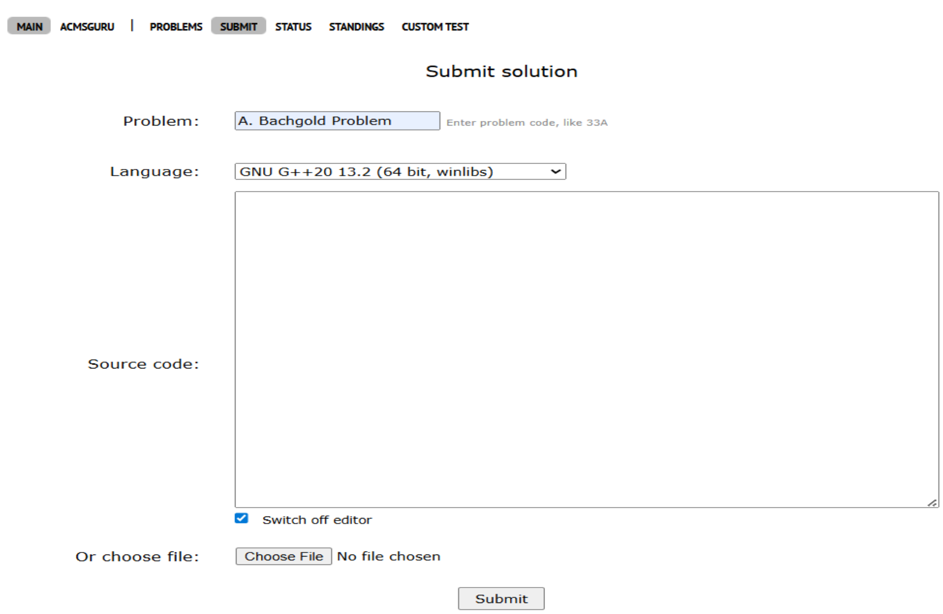




Compiler:

The website has a compiler it supports different types of programming languages such as C, C++, Java, Python, Ruby, C#, Rust, Javascript etc

Screenshot:



User Profile Page

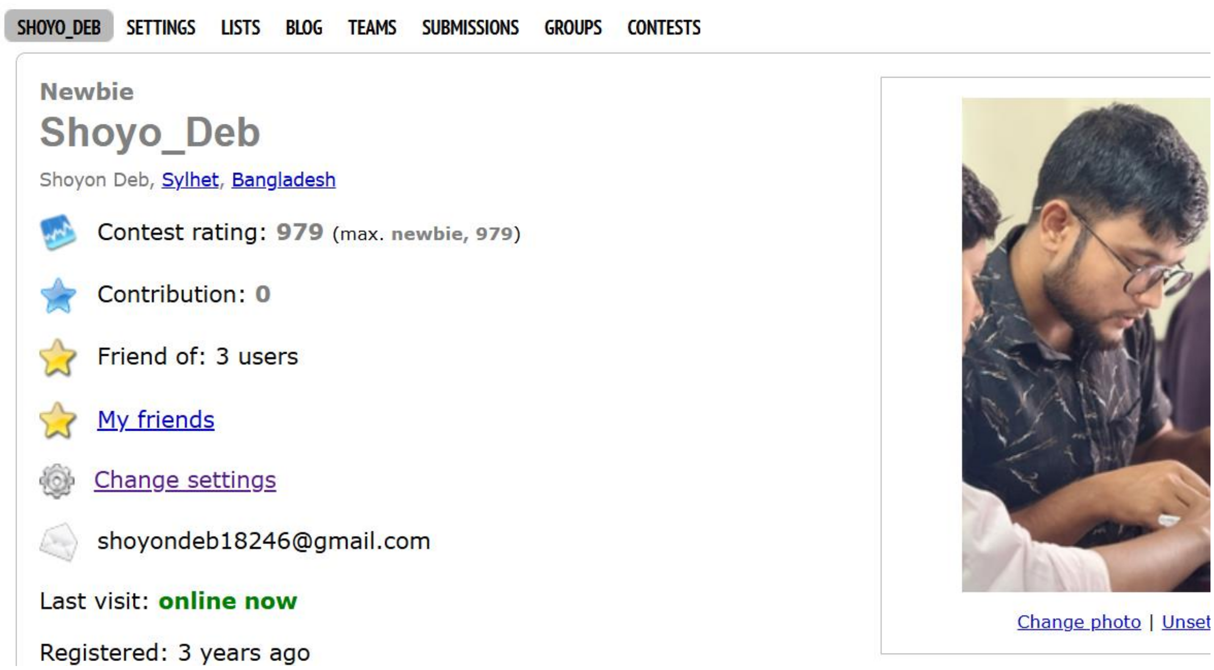
Description:

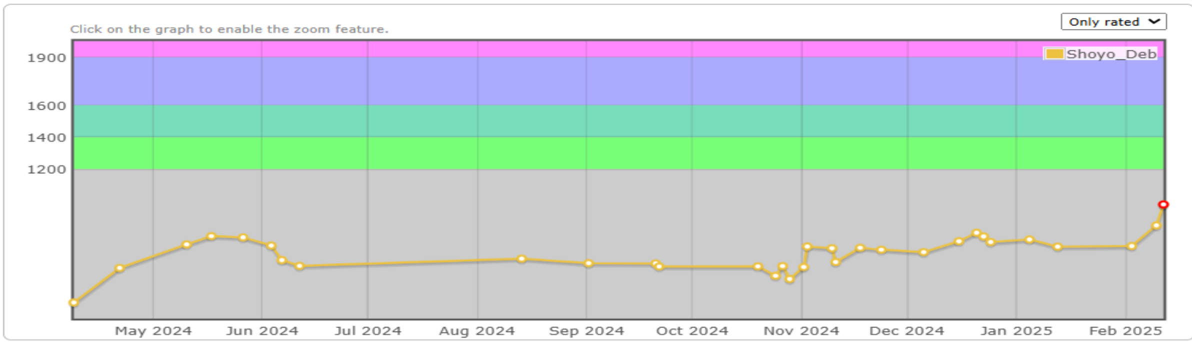
- Shows **rating, rank, recent contests, and solved problems**.
- Graphs track performance over time and gamify progress.

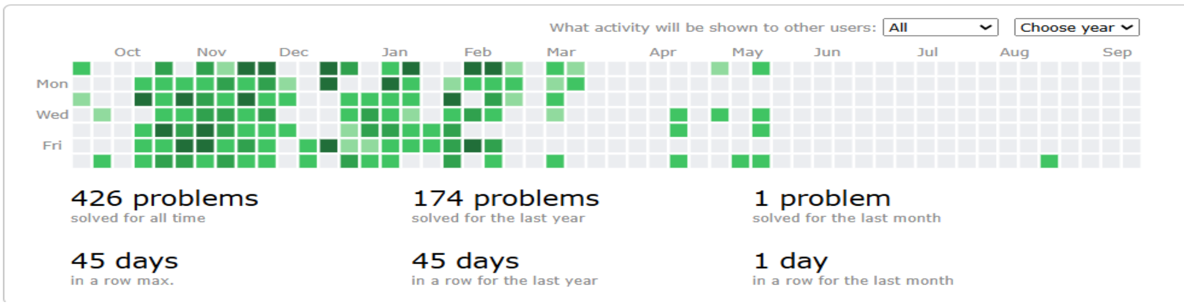
Observations:

- Motivates users via visible progress and achievements.
- Can be complex for beginners due to many stats and graphs.

Screenshot:



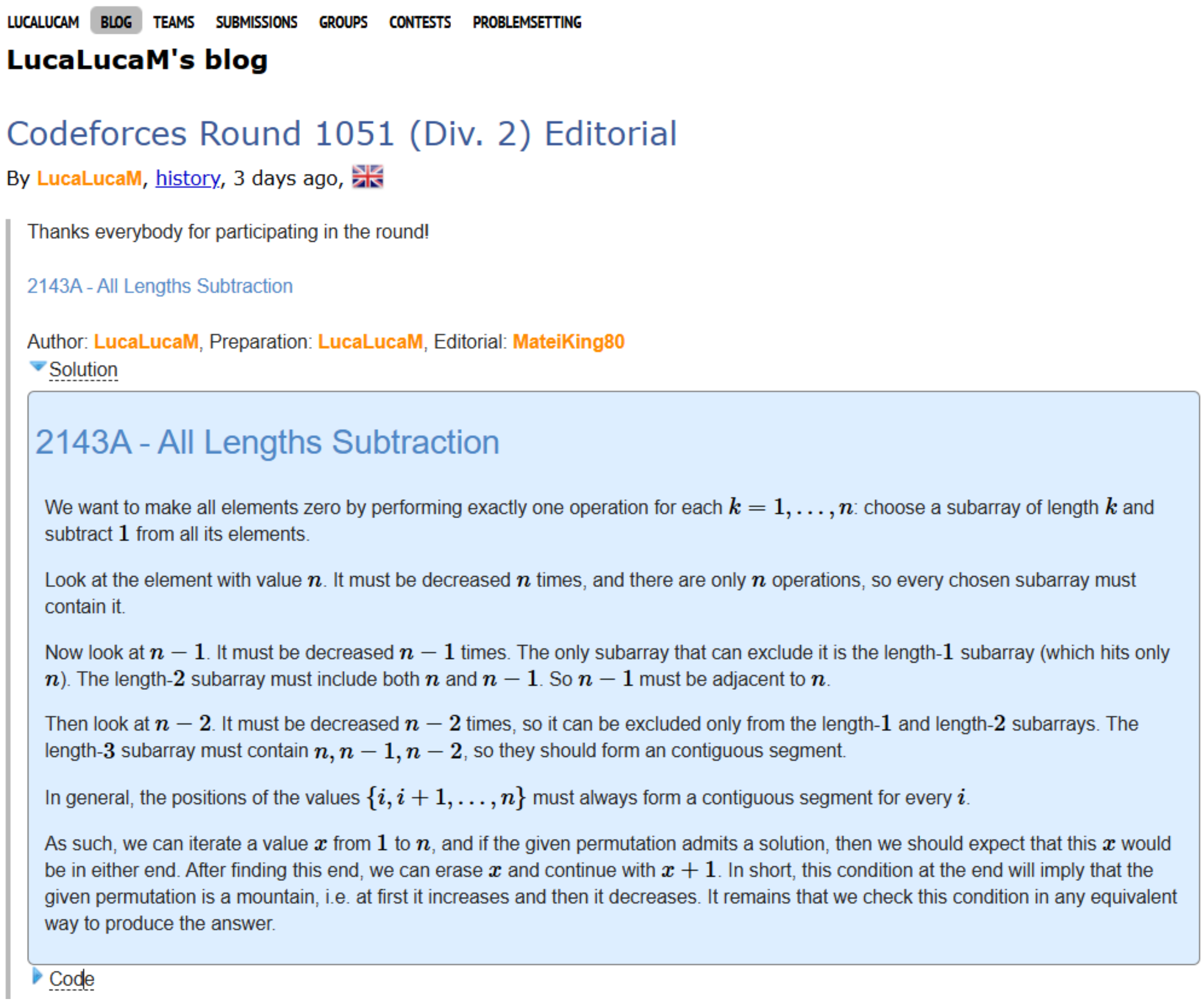




Rating System

- **Elo-Based Rating**: Codeforces employs an Elo-based rating system, similar to that used in chess, to assess a participant's skill level. Ratings are updated after each contest based on performance relative to other participants.
- **Titles and Divisions**: Users are assigned titles such as "Newbie," "Pupil," "Specialist," etc., based on their ratings. These titles serve as milestones and motivators for users.

Each of the practice problem has Tutorial:



LUCALUCAM BLOG TEAMS SUBMISSIONS GROUPS CONTESTS PROBLEMSETTING

LucaLucaM's blog

Codeforces Round 1051 (Div. 2) Editorial

By LucaLucaM, history, 3 days ago,

Thanks everybody for participating in the round!

2143A - All Lengths Subtraction

Author: LucaLucaM, Preparation: LucaLucaM, Editorial: MateiKing80

Solution

2143A - All Lengths Subtraction

We want to make all elements zero by performing exactly one operation for each $k = 1, \ldots, n$: choose a subarray of length k and subtract 1 from all its elements.

Look at the element with value n. It must be decreased n times, and there are only n operations, so every chosen subarray must contain it.

Now look at $n - 1$. It must be decreased $n - 1$ times. The only subarray that can exclude it is the length-1 subarray (which hits only n). The length-2 subarray must include both n and $n - 1$. So $n - 1$ must be adjacent to n.

Then look at $n - 2$. It must be decreased $n - 2$ times, so it can be excluded only from the length-1 and length-2 subarrays. The length-3 subarray must contain $n, n - 1, n - 2$, so they should form an contiguous segment.

In general, the positions of the values $\{i, i + 1, \ldots, n\}$ must always form a contiguous segment for every i.

As such, we can iterate a value x from 1 to n, and if the given permutation admits a solution, then we should expect that this x would be in either end. After finding this end, we can erase x and continue with $x + 1$. In short, this condition at the end will imply that the given permutation is a mountain, i.e. at first it increases and then it decreases. It remains that we check this condition in any equivalent way to produce the answer.

Code

5. Community & Discussion

Description:

- Blogs, forums, and problem discussions create a **strong social environment**.
- Users can share tutorials, solutions, and participate in discussions.

Observations:

- Encourages learning from peers and expert coders.
- Some discussions can be technical and hard to follow for beginners.


SUMMERSKY BLOG TEAMS SUBMISSIONS CONTESTS

SummerSky's blog

Welcome to discuss problems

By **SummerSky**, 6 years ago,

I started practicing from round 1 and now I have completed about 150 rounds. Of course I did not solve all of the problems since some of them are too difficult for me...

Although the tutorials are very good, however I feel that sometimes it might be still a little bit difficult for beginners, like me, to fully understand, especially for many div1 problems. On the other hand, as the problems are quite interesting and we can learn many clever ideas and techniques behind them, I think discussion helps a lot for obtaining deeper understanding.

Therefore, if anyone is also trying to solve problems one round after another, you are welcome to discuss problems with me, especially for those problems that I have submitted and got accepted (well, after all, I don't know how to solve those that I have never submitted...)

I would like to share any ideas or techniques involved in each problem that I have solved. Hope that every one could reach 'red' someday :D

+44 SummerSky 6 years ago 17

Comments (15) Show archived | Write comment?

6 years ago, hide # | 0

6. Strengths

- Frequent contests and active community.
- Strong gamification via rating system.
- Large catalog of problems with tags and difficulty levels.
- Editorials and discussions enhance learning.

7. Improvement Points

- Complex scoring and division system may confuse beginners.
- UI can be overwhelming for new users.
- Beginners may need guided learning paths or hints for initial practice.
- Rating system should be more transparent to understand beginners.

*Admin Panel Functionalities

The admin panel is used by Codeforces staff and problem setters to manage the platform and contests. Main functionalities likely include:

User Management

- View and manage user accounts (banning spammers, resetting passwords, adjusting ratings in rare cases).
- Assign or manage roles (contest setter, problem tester, admin).

- Track suspicious behavior or cheating attempts.

Contest Management

- Create, schedule, and configure contests (Div.1, Div.2, combined).
- Set contest start/end times and manage timezones.
- Enable or disable the hacking phase and virtual participation.
- Assign problem sets to contests.

Problem Management

- Upload new problems, including:
  - Statement, constraints, examples, tags, difficulty rating.
  - Editorial content for solutions.
  - Test cases for judging (correct, edge, and stress tests).
- Edit or archive old problems.
- Categorize problems (tags, ratings, contests).

Platform Maintenance

- Monitor server performance, uptime, and errors.
- Manage announcements and news.
- Backup or restore the database (users, contests, submissions).

Analytics & Reports

- Monitor contest participation statistics.
- Track submission statistics (number of submissions, success rates, language usage).
- Track rating distribution and changes over time.

Judge Panel

The judge panel is the automated system that evaluates user submissions during contests or practice. Its functionalities include:

Submission Evaluation

- Receive and queue submissions in real time.
- Compile code for multiple programming languages (C++, Java, Python, etc.).
- Run the submission against hidden test cases (both sample and system tests).

Result Processing

- Determine verdicts:
  - Accepted (AC)
  - Wrong Answer (WA)
  - Time Limit Exceeded (TLE)
  - Memory Limit Exceeded (MLE)
  - Runtime Error (RE)
  - Compilation Error (CE)
- Store detailed results including failed test cases and execution time.

Contest Rules Enforcement

- Apply scoring rules (partial scores, time-based penalties, hacks).
- Handle **virtual participation** (submissions are evaluated but do not affect ratings).
- Check for cheating or plagiarism (similar submissions, code similarity analysis).

Test Case Management

- Maintain problem-specific test case sets.
- Add, remove, or update test cases for new or revised problems.
- Classify test cases (sample vs system, normal vs stress tests).

Performance and Load Management

- Distribute judging across multiple servers to handle simultaneous submissions during contests.
- Monitor queue lengths, processing times, and system resource usage.